**Department of Computer Science**

# Summative Coursework Set Front Page

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|---|---|
| Module Title | Operating Systems and Computer Networking |
| Module Code | CS2ON |
| Lecturer responsible | Prof. Shuang-Hua-Yang and Nisha Singh |
| Type of Assignment<br>(e.g., technical report, set exercise, in-class test) | Technical Report |
| Individual or Group Assignment | Individual |
| Weighting of the Assignment | 30% |
| Word count/page limit | Up to 8 pages [Up to 4 pages- Part A and up to 4 pages- Part B] |
| Expected hrs spent for the assignment | 12 hours of independent work + lab sessions |
| Items to be submitted | One pdf document |
| Work to be submitted on-line via Blackboard Ultra by | 19-05-2025 |
| Work will be marked and returned by | After 15 working days |
| Artificial Intelligence Tools (select one of these) | May be used to support work |

**Note**

By submitting this work, you are certifying that you have read the assessment guidelines, which are displayed in the folder of Assessment on the Blackboard course for this module, and that you have conformed to and understand the associated policies and practices, including those on:

- Submitting your own work, not that of other people or systems (including those using artificial intelligence), and the associated penalties for Academic Misconduct
- Submitting by the specified deadline, and the penalties associated with late submission (if allowed)
- The exceptional circumstances system
- For students with relevant needs, attaching with a green sticker

# 1. Assessment classifications

## 1.1 Assessment classifications of Part A

| | |
|---|---|
| First Class (>= 70%) | Fully working programs with correct results show fully commented functions with suitably concise code, and the extra relevant functions, in a well-presented report |
| Upper Second (60-69%) | Three of the above done well |
| Lower Second (50-59%) | Two of the above |
| Third (40-49%) | One of the above |
| Pass (35-39%) | Basic work |
| Fail (0-34%) | Poor work |

## 1.2 Assessment classifications of Part B

| | |
|---|---|
| First Class (>= 70%) | An excellent implementation of Small Office Network. Impeccable configurations of DNS and HTTP servers. The report is well structured and narrated, including the implementation details of the ICMP protocol, results, and appraisal. |
| Upper Second (60-69%) | A good implementation of Small Office Network. DNS and HTTP servers are functional. The report is well structured and narrated, including the implementation details of the ICMP protocol, results, and appraisal. |
| Lower Second (50-59%) | A reasonable implementation of Small Office Network. The report is well structured and narrated, including the implementation details of the ICMP protocol, results, and appraisal. |
| Third (40-49%) | A good attempt at implementation of the Small Office Network. The report is well structured and narrated, including the implementation details of the ICMP protocol, results, and appraisal. |
| Pass (35-39%) | The report is readable, includes the implementation preparations of the ICMP protocol, issues, and appraisal. |
| Fail (0-34%) | No attempts or the provided implementation/report are not readable. |

## 2. Assignment description

The main objective of the coursework is to test the skills students have developed during the lab sessions, implementing concepts in operating systems and computer networking.

The assignment consists of two parts: part A for OS (Operating Systems), and part B for Computer Networking

The weight of the coursework is 30%. Students should submit a report to Blackboard by 19th May, whose contents are specified below.

Marks will be allotted based on the correctness of the answers, originality, readability, and organization of the submitted report. You may lose marks for coding-based questions even if your code is correct or there needs to be more comments/justifications.

**Resources Required:**

You should make use of these specific tools/ Operating systems and resources:

- Linux OS
- Packet Tracer
- Lecture/Lab Materials.
- Textbook: Computer Networking: a top-down approach and Operating System Concepts by Abraham Silberschatz-(9th,2012_12).

## 3. Assignment submission requirements

An electronic report in **PDF format** describing your work. The report should be a maximum of 8 pages (A4 sheets), at 12pt Times Roman font (including diagrams, code fragments, figures, and references). Screenshots should be included as appendix.

For part B, students must **upload the `.pkt` file to GitLab** and **include the link** in the PDF. The report must include:

### *a. Front page of the student's submission*

(*the following are compulsory)*

Module Code:

Assignment report Title: **Linux Operating System and Building a small Office Network**

Student Number (e.g. 25098635):

Actual hrs spent for the assignment:

Which Artificial Intelligence tools used (if applicable):

**b.** For Part A: Introduction to all the commands and justification of answers to be provided in the PDF report for part A and include all the screenshots in the appendix form.

**c.** For Part B: A small office network implementation .pkt file, network overview, and screenshots as appendix should be provided along with discussions.

**d.** Fill and include the self-assessment form (attached below)

Attempt the code-based questions by following the instructions given below:

- Do not insert code as images. Providing code as images makes marking the code difficult, and you may lose marks.
- Provide a well-commented, precise, working, and structured code.
- Clear and precise description of what is happening in the code.

## 4. Tasks for Part A

This section provides the tasks to be submitted in Part A of the PDF report.

**Task 1: Use Linux Commands for the following File and Directory Management tasks:**

Write a Bash script using Linux commands that perform the following tasks:

1. Create three directories with names that you generate based on a random or meaningful pattern (e.g., use your initials, a random string, or a combination of words).
2. Inside each directory, create two files. The file names must also be generated based on a unique pattern you choose (e.g., using your student ID, a random number, or a specific naming convention you define).
3. Add unique sample content to each file. The content should be descriptive and specific to the directory and file, such as:

    - The first file in each directory could describe something specific to that directory (e.g., Dir1/Task1.txt might describe a project you're working on).
    - The second file could contain additional details, such as a reference or explanation related to the directory.

4. Search for a specific keyword (e.g., "project" or "task") within all files in the directories and display the results.
5. List the contents of one directory with detailed information, showing permissions, file sizes, and ownership.
6. Copy the content of one file to another file and rename it.
7. Run strace on the script designed in task 1, with different filters (open, write, execve). Justify your answer.

Write the bash script and give brief information on each Linux command performed, in the PDF report and provide the corresponding screenshots in the appendix. **[20 marks]**

**Task 2: Use *strace* to look at the execution of the following program.** **[20 marks]**

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
int main(void) {
   int pfds[2];
   pipe(pfds);  // Create a pipe
   if (!fork()) {  // Child process
      close(1);  // Close stdout
      dup(pfds[1]);  // Redirect stdout to pipe write end
      close(pfds[0]);  // Close read end of the pipe
      execlp("cat", "cat", "file.txt", NULL);  // Execute 'cat file.txt'
```

```
  } else {  // Parent process
     close(0);  // Close stdin
     dup(pfds[0]);  // Redirect stdin to pipe read end
     close(pfds[1]);  // Close write end of the pipe
     execlp("grep", "grep", "pattern", NULL);  // Execute 'grep pattern'
  }
  return 0;
}
```

Trace the execution of the above program using strace. Check the man page of strace and filter the output to show only process management and pipe system calls. In your report answer the following:

i) Where is the pipe set up, and what file descriptors are created?
ii) Where are the new processes created, and what are their process IDs?
iii) Where are the commands cat file.txt and grep pattern executed, indicating which process is responsible for each?
iv) Explain the role of dup() and close() system calls in the above program. How do they affect the process's file descriptor table and input/output redirection?

Along with screenshots of execution, justify your answer for all the parts in brief.

**Task 3: Consider the c program and answer the following questions: [10 marks]**

```
#include <stdio.h>
#include <unistd.h>

int main() {
printf("Parent process: PID = %d\n", getpid());

fork();

if (fork())
{
  fork();
}
else
{
  fork();
  fork();
}
printf("Process: PID = %d, Parent PID = %d\n", getpid(),   getppid());

   return 0;
}
```

i. Analyze and explain the behaviour of this program, including the role of the fork() system call within the provided ”if-else” block.
ii. Draw a process tree representing the hierarchy of parent-child relationships.
iii. Explain the process creation step by step, analyzing how many processes exist at each fork() statement.

## 5. Tasks for Part B

This section provides the tasks to be submitted in Part B of the PDF report.

**Task 1:** Packet tracer is a simulation tool which helps to configure and visualise a network in logical and physical mode. The objective is to configure a small office Local Area Network (LAN) where 5 PCs and 1 printer are connected to a wireless router through wires. 1 mobile, 1 tablet PC are connected to the same wireless router which is acting as a DHCP server and default gateway provided by Internet Service Provider (ISP) to connect all those devices to the internet.

Configure those devices with the IP addresses

Default gateway- 192.168.1.1/24

Reserve IP addresses for those devices through DHCP

PC1- 192.168.1.10/24

PC2- 192.168.1.11/24

PC3- 192.168.1.12/24

PC4- 192.168.1.13/24

PC5- 192.168.1.14/24

PC6- 192.168.1.15/24

Printer – 192.168.1.253/24

Wireless Router configuration would be – SSID- pollyvacherwireless, with WPA2 Enterprise AES with shared key - P0llyVch3r. (Hint: you may need to set up a Radius AAA server). You then test at least two pairs of device-to-device connections using proper networking command(s) and check that different devices can communicate with each other.

**Task 2:** Install 1 DNS Server (192.168.1.2/24) and 1 HTTP server (192.168.1.3/24) into the previous office network. Update DHCP configuration to reserve IP addresses for the new servers. On the DNS Server, add an A record to create a mapping between www.pollyvacher.ac.uk and the HTTP server. Test the DNS resolution for www.pollyvacher.ac.uk using proper networking command(s) on at least 1 PC and 1 mobile device. Finally, use the web browser to open www.pollyvacher.ac.uk on at least 1 PC and 1 mobile device.

## Marking Scheme for Part A

The table below shows the mark scheme for individual tasks.

| Task# | Marks Distribution | Weigh-age out of 100 |
|---|---|---|
| 1 | i. Create three directories [3 marks], ii. Creation of files [2 marks], iii. Adding content to files [2 marks], IV. Search a specific keyword in files[ 4 marks], V. List content of directory [2 marks], Vi. Copy the content of one file to another file and rename it [3 marks] , Vii. Run 'strace' command with justification [4 marks]. | 20 marks |
| 2 | Execution of command 'strace' to show process management and pipe system calls with justification[4 marks] , i. Where is the pipe set up, and what file descriptors are created [4 marks], ii. Where are the new processes created, and what are their process IDs?[4 marks] , iii. Where are the commands cat file.txt and grep pattern executed, indicating which process is responsible for each?[4 marks] , iv. Explain the role of dup() and close() system calls in the program. How do they affect the process's file descriptor table and input/output redirection? [4 marks] | 20 marks |
| 3 | Explain the behaviour of the program [3 marks], Creation of process tree [ 4 marks], explanation of process creation [3 marks] | 10 marks |

## Marking Scheme for Part B

You must fill and submit the self-assessment form below with your submission.

| Section (# marks) | Students own assessment | Marks |
|---|---|---|
| Task 1. (20 marks) | **Building a Small Office Network** | (total 20) |
| | [ ] All devices connection | 4 |
| | [ ] All devices IP configuration | 4 |
| | [ ] DHCP server configuration | 4 |
| | [ ] Wireless router and AAA server setup | 4 |
| | [ ] Connectivity tests (x2) | 4 |
| Task 2. (20 marks) | **Adding DNS and HTTP Servers** | (total 20) |
| | [ ] All devices connection and IP configuration | 4 |
| | [ ] DNS Server configuration | 4 |
| | [ ] HTTP Server configuration | 4 |
| | [ ] DNS resolution tests (x2) | 4 |
| | [ ] HTTP service tests (x2) | 4 |
| Report. (10 marks) | **Quality of the submitted report** | (total 10) |
| | [ ] Introduction | 1 |
| | [ ] Small Office Network Setup with tests | 5 |
| | [ ] DNS and HTTP Servers Setup with tests | 3 |
| | [ ] Discussion, conclusion, and personal reflection | 1 |